

Navin Kumar

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SUMMARY

AI Engineer with 1+ year of experience building production RAG pipelines, tool-using LLM agents, and end-to-end AI-powered features on cloud platforms (GCP, AWS). Proficient in Python, with hands-on delivery of embeddings and vector search, LLM orchestration, structured outputs, prompt engineering, context construction, and evaluation of model output. Track record of shipping AI capabilities into customer-facing products in fast-moving environments with strong ownership.

EDUCATION

Indian Institute of Information Technology (IIIT) Vadodara

Dec 2021 – May 2025

Bachelor of Technology in Computer Science & Engineering

EXPERIENCE

AI Engineer — DigitalxCode

May 2025 - Present

Stock Analysis Agent

- Built an agentic stock intelligence platform using GPT-4o and a custom orchestration layer with four tool/function-calling agents bucket generation, RAG context retrieval, LLM health validation, and DB write autonomously producing investment signals across 24 equity buckets daily with zero human intervention.
- Implemented a RAG pipeline with ChromaDB vector database and OpenAI `text-embedding-3-small` to chunk and embed daily stock outputs, managing tokens and context windows to enable the agent to reason about portfolio drift via embeddings and vector search over historical data.
- Deployed a containerised system on GCP via Docker Compose orchestrating FastAPI (Python), Next.js, PostgreSQL, Redis, and ChromaDB across 5 services with automated nightly LLM generation jobs on a production dashboard.

Drone Perception & Face Recognition

- Fine-tuned RT-DETR transformer using PyTorch and Hugging Face on VisDrone (10 classes) to achieve 90%+ mAP across 5,000+ frames; built evaluation and prompt-testing pipelines tracking per-class precision/recall and latency.
- Reduced inference latency to under 33ms (30+ FPS) using model optimization and runtime profiling to deploy real-time drone perception alongside an InsightFace + OpenCV face recognition pipeline on live video.
- Performed model failure analysis and evaluation of model output on misclassified samples using confusion-matrix diagnostics to drive targeted dataset improvements and retraining cycles.

SDE Intern — EaseMyGrowth

Jan 2025 - May 2025

Time-Series Forecasting Service

- Owned end-to-end deployment of a time-series forecasting service using Docker and GCP to deliver model selection, SQL-driven data pipelines (10,000+ rows of document-heavy data), and integration into a client-facing dashboard.
- Integrated LLM-based workflows into downstream analytics using structured outputs, prompt engineering, and context construction to produce actionable business intelligence.
- Designed cloud-agnostic Docker workflows compatible with both GCP and AWS (ECS/EKS) to ensure cross-environment deployment consistency.

PROJECTS

AgentFuse — Logical Circuit Breaker for Autonomous Agents

[GitHub](#)

- Built a framework-agnostic Python supervisor using OpenAI AgentKit (with OpenAI SDK, LangChain/LangGraph support) that monitors every tool call, route, and token in real time to detect agent failure modes looping, goal drift, logic traps, and token burn.
- Implemented automatic steering recovery using OpenAI o4-mini reasoning model with embeddings-based drift detection to generate corrections on trip, injecting them back into the agent or escalating to a human when recovery isn't safe.
- Shipped a live observability dashboard (JS + SVG on GitHub Pages) with JSONL traces and rich terminal output to replay every trip and recovery for full agent run transparency.

Human Activity Recognition in Dark

[GitHub](#)

- Built an LRCN architecture (CNN + LSTM) for activity recognition in low-light video; improved accuracy from 72% to 91% through targeted preprocessing and model tuning; validated with loss curves and confusion-matrix evaluation.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C/C++, Node.js

AI / ML: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, OpenCV

LLM Orchestration & RAG: OpenAI SDK, OpenAI AgentKit, LangChain, RAG, ChromaDB, pgvector, Pinecone, embeddings & vector search, prompt engineering, structured outputs, tool calling, context windows, evaluation pipelines

Infra & Production: Docker, Docker Compose, Google Cloud Platform, AWS (ECS/EKS), FastAPI, React, PostgreSQL, Redis, Git/GitHub, REST APIs